

BrainLift for AlphaBrilliant: Koji, the Voice-First AI Tutor, and the Learning-Science Layer That Makes It Stick

Owners

Khoi Lam (Primary Owner)

Purpose

Primary Purpose: To articulate the pedagogical and architectural thesis behind **AlphaBrilliant** (a Brilliant.org-style, learn-by-doing course on the Pythagorean theorem). The app is built on a deterministic truth engine, adds an additive, voice-first AI tutor (**Koji**) on top of it, and wraps both in a learning-science retention layer that makes what it teaches last. Its thesis is that the right way to put a large language model into a *teaching* product is **voice-first, answer-withholding, and architecturally subordinate to the deterministic engine**, and that the app teaches completely with AI off. Koji can *talk and listen, see the lesson's typed state, and operate the app* via tools, but he is never allowed to be the source of mathematical truth.

Making it stick. The same thesis runs from *teaching a concept once* to *making it last*. The app layers four evidence-based learning-science techniques onto the very same deterministic engine: **spaced repetition** (a faithful TypeScript port of **FSRS-6**, the modern scheduler Anki now ships, *not* legacy SM-2), **retrieval practice**, **durable mastery learning**, and **sharpened, AI-off explanatory feedback**, and holds every one of them to the same two rules the rest of the app lives by: each is **additive** (it rides the existing primitives and adds *no* AI dependency) and **AI-off-safe** (it works byte-for-byte with the model switched off). The defining bet of this layer is a redefinition: **mastery is not a one-time pass; it is *survival of a spaced review***. The retention layer therefore fuses the mastery gate with the forgetting curve instead of treating spacing, mastery, and feedback as three separate features, delivering spaced repetition, mastery scheduling, and mastery-driven sequencing as one coherent design (see Appendix A.3).

This document captures the *thinking* behind the product: the spiky points of view, the learning science that grounds them, and the experts whose work shaped them, not the build log (the *making* lives in Appendices A and B).

In Scope

- The **pedagogy of an AI math tutor**: when to help, how to help, and (most importantly) when to *refuse* to help.
- The case for **voice as the primary learning modality** for an at-home consumer learning app, and the cognitive science that both supports and complicates it.
- The **reliability architecture** that lets an unreliable, hallucination-prone model power a product where a single wrong answer is disqualifying (the "verification firewall").
- **Grounding / context engineering**: feeding the model typed lesson state rather than scraped screen text.
- **Additivity**: why "AI-native" must be a leverage claim, not a dependency claim.
- **Model-authored, adaptive problem generation ("Infinite Practice")** as a deliberate, *scoped exception* to the firewall: a strong reasoning model authors and owns the correctness of supplemental practice, while the firewall stays absolute for the core teaching engine (see SPOV 2).
- **Spaced repetition as the spine of retention**: an FSRS-6 scheduler applied *per skill*, a home/course-map **"Reviews due (N)"** hub with a due-soon forecast, and a one-tap review session that resurfaces wrong/assisted skills soonest.
- **Retrieval practice over recognition**: spaced recall, a *cumulative* level review, an opening recall warm-up before new lessons, and reviews that favor *generative* inputs with the scaffold dropped.
- **Durable, gated mastery**: a per-skill **mastery meter**, the "mastery = survives a spaced review" rule, a Bloom-style corrective loop (formative → corrective → re-test), and a level gate that locks the next level until its skills are mastered.
- **Explanatory feedback that teaches with AI off**: the deterministic misconception classifier ported *out* of the AI path into the hand-authored static feedback, so wrong answers teach even with the model off.
- A small **skill taxonomy (~7 skills)** that underpins the spacing, retrieval, and mastery layers.

Out of Scope

- Building a general-purpose chatbot or open-ended conversational agent. Koji is grounded and tool-scoped, not a chat box.
 - AI-authored **pedagogy**: new lessons, new interaction kinds, AI-generated figures, or AI-rewritten feedback. Generation is restricted to *problems within the proven schema*.
 - **Per-learner ML training and open-ended adaptive sequencing**. The app delivers real learning-science scheduling (spaced repetition, durable mastery, mastery-driven next steps), but deliberately stops short of three things: **training FSRs's parameters per learner** (we ship the published defaults, see SPOV 7), **interleaving as a headline feature** (it is folded into mixed/cumulative review sessions, not sold as its own mode), and **full cross-lesson adaptive sequencing** beyond what mastery and the scheduler already drive. The fixed, hand-authored lesson path still defines *what* is taught and in what order.
 - Subjects beyond the Pythagorean theorem; multi-learner / classroom features; a parent/teacher analytics dashboard.
 - Claims about voice that this build does not yet earn, e.g. measured retention lifts. Where the science is contested, this document says so rather than overclaiming.
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DOK 4: Spiky Points of View (SPOVs)

Each Spiky POV names the conventional view, then the contrarian stance that overturns it. Every one below is a defensible position, synthesized from the learning science in the Knowledge Tree and grounded in how the product was actually built.

SPOV 1: The study-science default treats serious learning as a product of effortful *writing*. For an at-home learning app, though, voice is the superior primary modality, because the binding constraint is engagement, not per-rep encoding depth, and a learner reasoning *aloud* is doing real encoding anyway.

The consensus counter-view, stated honestly and at full strength. The study-science default would pick text, and it has good reasons. Robert Bjork's **desirable difficulties** framework holds that conditions which *slow you down* during learning, making you work harder to produce and retrieve, are exactly what produce durable long-term retention (Bjork, 1994; Soderstrom & Bjork, 2015). The **generation effect** (Slamecka & Graf, 1978) shows that information a learner *generates* is remembered better than

information merely read, and typing forces generation. Writing is slow, effortful, and legible to the writer's own metacognition; it is the canonical "desirable difficulty." A serious learning scientist, handed the choice between "the student types out their reasoning" and "the student chats with a friendly voice," would, all else equal, pick the keyboard. Three of this document's own experts lean this way: Justin Sung warns that the moment a learner stops actively constructing and starts passively listening, encoding quality collapses; Mayer's **modality principle** has a documented boundary condition: printed words can beat spoken words precisely when the material is *symbol-heavy and technical* (Mayer, 2005), and mathematics is nothing if not symbol-heavy; and the production-effect literature is explicit that producing information aloud improves *retention and distinctiveness* but **does not** improve deep comprehension or inference (MacLeod & Bodner, 2017). On the merits of a single, controlled study rep, text has the stronger hand.

The spiky case for voice-first. That entire argument quietly assumes the realistic alternative to "talking to Koji" is "carefully writing an essay-length chain of reasoning." For a consumer app a teenager uses on a phone, on a couch, on a Tuesday night, **it is not**. The realistic alternative to talking is *quitting*: closing the app. The binding constraint for at-home learning is not the encoding depth of any given repetition; it is **whether the learner takes the next repetition at all**. Voice has the lowest activation energy of any input modality: no typing math notation on a touch keyboard, no friction, no "ugh, later." It keeps the learner *inside the productive-struggle loop* long enough for the loop to matter. A slightly shallower rep that actually happens beats a textbook-perfect rep that doesn't. This is the same logic Bloom's 2-sigma problem rests on (the gains come from *staying in* a tight feedback loop with a responsive tutor) and the same logic the Freeman et al. (2014) active-learning meta-analysis rests on: the intervention only works if students keep participating. Voice is also not as cognitively cheap as the consensus assumes; when designed correctly, it is itself a strong encoding act:

- **The production effect.** Saying material *aloud* reliably beats reading it silently, an effect named and characterized by MacLeod, Gopie, Hourihan, Neary & Ozubko (2010). The mechanism is *distinctiveness*: a spoken record is more discriminable at retrieval. Crucially, the benefit accrues to **the person doing the producing**. So when the design forces the *learner*, not Koji, to talk through their reasoning, voice converts from a passive channel into an active encoding event.

- **Mayer's modality principle.** People learn more deeply from graphics + spoken words than from graphics + on-screen text, because audio offloads the words onto the verbal channel and frees the visual channel for the figure (Mayer, 2005; modality effect median $d \approx 1.0$). Koji speaks while the triangle and the KaTeX equation stay on screen: eyes on the problem, words in the ear. The visual channel is never asked to read prose *and* parse a diagram at once.
- **Sung's active-encoding lens.** Voice aligns with high-quality encoding *if and only if* it is engineered so the learner reasons aloud rather than passively absorbing Koji's explanation.

Why this is a strong SPOV and not a hot take: the failure mode is real, and the product is built to defeat it. Voice's obvious failure mode is degenerating into passive listening: Koji explains, the learner nods, nothing is encoded. The science above is precisely what tells us how to avoid it, and the build implements those mitigations as code, not vibes:

- Koji is forbidden from stating answers. His realtime persona reads, verbatim: "*Never say the final numeric answer yourself, and never compute it out loud*". The correct answer is deliberately **omitted from the voice prompt entirely**: the grounding step that feeds Koji the lesson state is built from the givens and the learner's own answer, never from the correct answer.
- The reveal is **effort-gated**, and one of the two ways to satisfy the gate is *talking it through*: the gate checks whether the learner has talked to Koji. The design literally rewards the learner for producing speech, which is the production effect operationalized.
- Koji's instructions tell him to "*Celebrate effort, not just correct answers*" and to nudge, not lecture.

And the honest scope: voice is an **engagement-and-retention** lever, not a comprehension silver bullet. The production effect doesn't deepen understanding on its own, so comprehension still rides on the *active reasoning* Koji elicits, the on-screen visuals, and the verified engine underneath. Voice-first is the right *primary* bet; text stays available in the same unified thread for the learner who wants to slow down and write.

SPOV 2: The reflex for making an LLM safe in a teaching product is better prompt engineering. In fact, for anything checkable, prompting is the wrong layer: you make the model *irrelevant to correctness* with a deterministic verification firewall, and demote the LLM to a phrasing engine.

The industry reflex when an AI says something wrong is to write a better prompt: "Be accurate." "Do not hallucinate." "Double-check your math." This is fundamentally unsound for a *teaching* product, where the cost function is asymmetric and brutal: a tutor that confidently emits a wrong Pythagorean answer even 1% of the time is not 99% good: it is disqualified, because it teaches the wrong thing with authority. And generative models are structurally bad at exactly this: as Khan Academy's own engineers put it, generative AI "was not designed to handle mathematical reasoning... it uses all of the language it is trained on to predict the most probable numbers... The most probable number in the training data is not always the correct answer" (Khan Academy, 2024).

So AlphaBrilliant refuses to negotiate with the model about truth wherever truth is *taught*. The architecture draws a hard line around everything the app presents as correct: **a pure, deterministic engine owns correctness, and nothing AI-proposed reaches the learner as taught truth until that engine certifies it.** That line is absolute for the core teaching engine; it is deliberately and narrowly relaxed for *supplemental* generated practice, the one scoped exception argued at the end of this SPOV.

- Ground truth is the deterministic grading engine, plus an independent math check for recomputation.
- **Generated *practice* problems are the one deliberate, scoped exception to this list** (full rationale at the end of this SPOV and in Appendix B.7). A strong reasoning model now authors the complete problem and *owns* its correctness, with every authoring constraint (whole-number answers, correct math, variety, figure rules) living in the **prompt**. The old deterministic round-trip that regraded each generated answer and independently re-derived its numerics (e.g. $\sqrt{a^2+b^2}$), *failing closed* on anything it could not recompute, has been **removed** for this path; only crash-prevention shape-validation and a deterministic fallback remain. This is the single place the model is trusted on correctness, and only because the stakes are low: a lone supplemental practice item, never anything the engine teaches as true.
- Even "reveal the answer" is **engine-computed**, never model-generated.

- Hints get a second backstop: the server post-checks the model's hint text for the rendered answer and swaps in a static hint if it leaks: a server-side answer-leak check plus a client-side answer-leak check. Defense in depth.

This is not anti-AI; it is the only design that makes the AI *safe to be additive*. The model still does what models are uniquely good at: warm, personalized, context-aware *phrasing*. It just never gets to decide what's true. Khanmigo arrived at the same conclusion independently, bolting a behind-the-scenes "math agent" onto GPT-4 to verify calculations against verified content (Khan Academy, 2024): strong convergent evidence that *demoting the model below a deterministic checker* is the correct architecture for math tutoring, not a quirk of this build.

The firewall is now *scoped*, not absolute, and the scope line is drawn exactly where truth is *taught*. For the core teaching engine, the hand-authored lessons whose answers the app presents *as correct*, everything above stays absolute and non-negotiable: a wrong taught answer mis-teaches with authority, which is the disqualifying, asymmetric cost this whole SPOV exists to defeat, and that argument stands at full strength. But *supplemental*, generated **Infinite Practice** is a different bet, taken deliberately: there the app trusts a strong reasoning model, constrained entirely by its prompt, to author correct problems, and it has removed the deterministic round-trip that used to re-derive every generated variant. The reasoning is a cost asymmetry, not a loss of nerve. A single wrong *practice* problem costs the learner far less than a single wrong *lesson*, and deterministic verification of every variant proved genuinely too rigid: to survive the round-trip the generator kept retreating to the same handful of memorized Pythagorean triples, and the integer gate rejected the near-misses, so "infinite" practice collapsed toward a dozen reskinned triples, the precise opposite of the variety that makes practice worth doing. So on the practice path *only*, the app accepts a small, honestly-stated risk, an occasional generated problem can now be wrong, by design, in exchange for the variety and reach that supplemental drilling needs. The firewall did not weaken; its scope sharpened to the place where correctness is load-bearing (see Appendix B.7).

SPOV 3: It is taken for granted that the best AI tutor is the most *helpful* one: fast, complete answers on demand. In reality, the most pedagogically valuable thing an AI tutor can do is *refuse to give the answer*: making the answer *earned* protects the retrieval and productive-struggle loop where learning actually happens.

An LLM's default temperament is sycophantic completion: ask for the answer, get the answer. For a search engine that's the product; for a tutor it is **actively harmful**, because it short-circuits the single most evidence-backed mechanism in all of learning science.

Roediger & Karpicke (2006) and Karpicke & Roediger (2008) show that *retrieval practice*, the effortful act of producing an answer from memory, produces dramatically better long-term retention than restudying, and, damningly, that students **cannot feel this working** (their confidence is uncorrelated with their actual retention). A tutor that hands over answers feels great and teaches little. It optimizes the learner's *comfort* against the learner's *learning*.

So Koji is built to withhold, and the withholding is enforced structurally, not just requested in a prompt:

- **Effort-gated.** The reveal tool refuses unless the learner has made a genuine attempt **and** engaged Koji (a hint or a conversation). No effort, no answer. The old instant "See answer" button is removed while AI is on.
- **Learner-initiated only.** Koji's persona forbids auto-revealing; the gate is the enforced backstop.
- **Personalized to the gap, not just the number.** When the reveal is earned, it names the learner's specific misconception, diagnosed deterministically (recognizing "added the legs," "forgot the square root," "subtracted the legs," "counted one square," "picked a leg, not the hypotenuse," ...), then walks them through the engine-computed working.
- **Marked as assisted.** A revealed step never counts as first-try mastery, so the learner's progress signal stays honest.

This is the practitioner consensus among people who actually ship tutors: Khanmigo's headline design choice is that, "*unlike other AI tools such as ChatGPT, Khanmigo doesn't just give answers... it guides learners to find the answer themselves,*" and Khan Academy tracks "giving the answer away before a student submitted" as a *guardrail metric to minimize* (Khan Academy, 2024). Restraint is the feature.

SPOV 4: The fashionable reading of "AI-native" is that the AI *is* the product. But "AI-native" must mean *additive, never load-bearing*: you build the AI-off path *first*, so that with the flag off the app is byte-for-byte the complete deterministic engine and the AI can never break the product.

"AI-native" is fashionable and usually means "the app falls over if the model is down, slow, rate-limited, or having a bad day." That is a *dependency* claim dressed up as an innovation claim, and it is a liability for a product whose job is to reliably teach. The spiky inversion: **AI-native is a *leverage* claim, not a *dependency* claim.** The AI should multiply the value of a product that already works without it.

AlphaBrilliant enforces this by construction:

- A single AI feature flag gates everything. With it off (the default), AlphaBrilliant behaves *byte-for-byte* like the complete deterministic engine: hand-written hints, hand-authored feedback, the fixed lesson path, zero AI, zero network calls to AI endpoints.
- Every client wrapper **early-returns a safe empty result** when the AI feature flag is off, so no AI code path can even execute. The Koji panel is gated at the host so it *does not exist* with AI off.
- On *any* runtime error or timeout, AI paths degrade gracefully to the deterministic static hints/feedback. The learner never hits a broken tutor or a dead end.
- The hand-written feedback isn't legacy cruft; it is *deliberately retained as the fallback*, which is why AI-rewriting it is explicitly out of scope.

Building the off-path first is what makes the firewall (SPOV 2) and the restraint (SPOV 3) cheap to guarantee: there is always a correct, hand-authored thing to fall back to. AI as the upgrade, never the foundation.

SPOV 5: The assumed lever for better AI output is a cleverer prompt. In practice, context engineering beats prompt engineering: ground the model in *typed lesson state*, never scraped DOM or rendered KaTeX.

Most "AI tutor" integrations feed the model the rendered page (the HTML, the KaTeX, the screen text) and then try to coax good behavior with prompt incantations. This is backwards. The rendered page is lossy, noisy, ambiguous, and expensive in tokens; a rendered snippet of KaTeX tells the model far less, far less reliably, than the structured object that *generated* it. The higher-leverage move is **context engineering**: hand the model the same typed state the app's own logic runs on.

In the AI layer:

- The grounding step that feeds Koji the lesson state constructs a compact, JSON-safe payload from the structured lesson state, the learner's typed answer, and their attempt history, **never** from rendered DOM, KaTeX, or screen text.
- The payload carries the engine-computed correct answer so the model never has to *compute* a fact (it only phrases around one), and a human-readable rendering of every answer so the server's leak-check has a real target.
- Both the text agent and the realtime voice agent are grounded from this one typed payload, so all surfaces see the same clean state.

Smaller prompts, lower cost, fewer hallucinations, and (because the givens are typed) the diagnosis and verification layers can reason about the same objects the model sees. Prompt engineering tunes *how* the model talks; context engineering controls *what it knows*, and what it knows is what makes it correct.

SPOVs 6–9: making it stick. The five above are about putting AI into a teaching product safely; the four below are about *retention*: how to make what's taught survive past the lesson. They ride the same architecture (additive, AI-off-safe, deterministic where it counts), and so they sharpen, rather than complicate, the thesis.

SPOV 6: Mastery is conventionally scored as *completion*: pass the check, unlock the next thing. In reality, a one-time pass isn't mastery; mastery is *surviving a spaced review*, so the gate must key off durable, spaced success, not a single correct attempt.

The consensus counter-view, stated honestly and at full strength. Mastery learning's standard operationalization is a high-bar check at the end of a unit (commonly 80–90%) before advancing, and that is already a real improvement over time-based promotion: Bloom's mastery learning is worth roughly +1 SD on its own (Bloom, 1984). The natural, near-universal way to *measure* that bar, the way most apps and even Bloom's own corrective loop do it, is an **immediate** post-test: answer correctly now, and you've shown mastery. It is simple, legible, and instantly motivating, and it avoids punishing a learner who has, in fact, just understood the idea.

The spiky case for durable mastery. An immediate pass measures *performance*, not *learning*, and the two routinely diverge: Soderstrom & Bjork (2015) draw exactly this line, and the spacing effect (Cepeda, Pashler, Vul, Wixted & Rohrer, 2006) is the proof: a fact you can produce today can be gone in a week. A mastery gate scored on the immediate attempt is therefore measuring the wrong variable: it certifies that the skill was *available a moment*

after it was taught, which is nearly guaranteed and nearly worthless as a retention signal. So the app redefines the bar: a skill counts as mastered only once it is produced correctly **after a spacing delay**, once it survives at least one scheduled review. This fuses two techniques the literature treats separately (mastery learning and spacing) into a single rule, and it is enforced structurally:

- A visible **per-skill mastery meter** gives the "clear mastery signal" the assignment (and Bloom) calls for, and it deliberately distinguishes **provisional** (passed immediately) from **mastered** (survived a spaced review), so the signal can't be gamed by a single lucky attempt.
- The gate requires *both* a first-try-correct demonstration *and* survival of ≥ 1 spaced FSRS review (SPOV 7). Mastery is therefore hard to *earn*, but once earned it is treated as a permanent achievement: a later lapse still reschedules the skill for reinforcement (its FSRS stability collapses, so it resurfaces sooner), but it never strips the badge, so one slip can't erase hard-won progress and demoralize the learner.
- The permanence is a deliberate *decoupling* of the retention *mechanism* (FSRS, which still resurfaces a lapsed skill) from the achievement *signal* (the badge, an earned marker rather than a live retention readout); the full case is argued in Appendix B.6.
- A **corrective loop** runs Bloom's formative $\rightarrow$ corrective $\rightarrow$ re-test cycle: a weak or lapsed skill routes to targeted Infinite-Practice on *that* skill, then is re-assessed on review rather than simply repeated.
- The next level and the **level review lock** until the level's skills are durably mastered. This is a deliberate reversal of the engine's original design (which had *no* sequential locking, every lesson was freely accessible); the learning-science layer adds the gate **only at level boundaries**, not per lesson, so exploration within a level is preserved while the *promotion* between levels is earned.

This is Bloom honored more faithfully than the usual implementation, with Bjork's learning-vs-performance distinction supplying the correction: the gate certifies durable learning, not a moment of performance.

SPOV 7: The intuitive way to review is "come back to it when you feel like it", or a fixed Monday/Wednesday/Friday cadence. In fact, an evidence-based scheduler that models the forgetting curve beats both: resurface each skill as predicted recall decays to a target retention, and resurface *missed* skills soonest. We are honest that we ship FSRs's *default* weights, not per-learner-trained ones.

The consensus counter-view, stated honestly and at full strength. Any spacing beats cramming, and you don't need much machinery to get most of the benefit: a Leitner box or legacy **SM-2** (an ease-factor $\times$ interval-multiplier) captures the bulk of the spacing effect with a few lines of code. For a course with only ~ 7 skills, reaching for a 21-parameter, benchmark-tuned scheduler looks like textbook over-engineering, the kind of thing the project's own complexity passes exist to cut. On that framing, the marginal accuracy of FSRs over a Leitner box on seven items is genuinely small, and the simpler tool is the responsible default.

The spiky case for a modeled scheduler. The win isn't squeezing the last few percent of accuracy out of seven skills; it's **correctness of the model** and a **clean mapping from our outcomes to memory updates**. FSRs-6 models each item's **Difficulty, Stability, and Retrievability**, fits a **power-law forgetting curve**, and schedules each skill to resurface when predicted retrievability decays to a **target retention of 0.9**, so "when to review" is a *prediction from a memory model*, not a hand-tuned multiplier with no notion of recall (which is precisely what SM-2 is). And it answers the one question we care about most with no hand-waving: a skill the learner **couldn't produce unaided**, wrong, or revealed via Koji's assisted path, is graded a **lapse** (FSRs's "Again"), which collapses its stability and brings it back *soon*. That is the assignment's "resurface ones a learner got wrong sooner," derived from the model rather than bolted on. It is enforced in code:

- A faithful **TypeScript port of FSRs-6** with its published default weights (sourced from `ankitects/anki`, `open-spaced-repetition/fsrs-rs`, and the FSRs algorithm spec).
- Memory state (D/S/R) persisted **per skill** under `users/{uid}/...`, updated only through the single outcome chokepoint (`recordStep`), so AI-on and AI-off write the same truth.
- The outcome $\rightarrow$ grade map made explicit: **first-try correct = pass; wrong or assisted = lapse**.

- A **"Reviews due (N)"** hub + due-soon forecast + one-tap session; the learning-science layer also captures **Infinite-Practice misses**, which were previously dropped on the floor.
- **No "review early" escape hatch.** The home card shows the due-soon forecast but deliberately gives the learner no button to pull a scheduled review forward. This is the one spot where our own judgment overrides the convenience default: FSRS decides *when* a skill is ripe, and the product *enforces* that wait. A review only pays off once recall has decayed enough that retrieving the skill is effortful, so letting a learner review on demand while the item is still fresh would spend the repetition for almost no gain (see Category 7.1: the retrieval-effort and spacing research). Adapting Anki's scheduler here means more than porting it: we also remove the affordances (like review-on-demand) that would let a well-meaning learner defeat the spacing the model just computed.

The honesty that earns the SPOV. We ship **default** weights, not per-learner-trained ones, and say so plainly. The tradeoff is real and the direction is favorable: on the public benchmark (`srs-benchmark` , ~10,000 user collections / ~350M reviews) FSRS beats SM-2 for ~99.5% of users **even with defaults**, and per-user optimization only trims error further, but it needs a review history we won't have on day one and adds ML infrastructure we judged not yet worth it. We are equally honest about the benchmark's limits: SM-2 was never designed to emit probabilities (the benchmark adapts it to compare), and the widely-quoted "20–30% fewer reviews" figure comes from *simulation*, not a controlled classroom trial. So this BrainLift treats "FSRS beats SM-2" as a strong **directional** claim, the same posture it takes toward Bloom's 2-sigma, a north star, not a measured promise.

SPOV 8: A "review" is usually built to *feel* like review, re-read the summary, recognize the right option, get a confidence bump. But a review should force *retrieval*, not *recognition*: generative input, scaffold dropped, cumulative, because effortful recall (a desirable difficulty) is what actually rebuilds the memory.

The consensus counter-view, stated honestly and at full strength. Re-exposure does *something*, and recognition tasks (multiple-choice, "does this look right?") are cheap to build, fast to answer, and feel productive, which is not nothing, given SPOV 1's whole argument that the binding constraint is engagement. A friendly, low-friction recognition review keeps the learner coming back, and a learner who reviews easily beats one who doesn't review at all.

The spiky case for retrieval over recognition. Recognition lets the learner coast on familiarity without ever *producing* the answer, and producing the answer is the entire mechanism: Roediger & Karpicke (2006) show retrieval practice trounces restudy for durable retention, and, damningly, learners are *blind* to it, so left to choose they will pick the comfortable recognition review that teaches least. Bjork's **desirable difficulties** says the harder recall is the point. So the app's reviews are engineered to be generative and unscaffolded, and the system makes that choice on the learner's behalf:

- Review item selection **favors generative interaction kinds**, e.g., a numeric "find c" over a multiple-choice "which value is c?" for the same skill, so the answer is *produced*, not spotted.
- The **scaffold is dropped**: the hints, worked structure, and visual aids that supported first acquisition are faded on review (scaffolding → desirable difficulty), so the skill stands on its own.
- The **level review is cumulative**, it interleaves *all* of the level's skills, so the learner must *select* the right approach, not merely repeat the last one drilled. (This is where interleaving lives: folded into the review, not sold as a headline mode.)
- An **opening recall warm-up** pulls the prerequisite skill from memory *before* a new lesson builds on it.

The honest bound. This raises difficulty on purpose, and a desirable difficulty taken too far becomes a discouraging one, exactly the cognitive-load boundary Mayer and the desirable-difficulties literature warn about. So the harder reviews keep their safety nets: the corrective loop (SPOV 6) catches a skill that's slipping, and Koji (AI on) is there to scaffold a learner who's stuck, the difficulty is *desirable*, not punishing.

SPOV 9: The reflex is that the AI gives the smartest feedback, a model reads the wrong answer and explains it. But the sharpest explanatory feedback belongs in the deterministic layer, not the model. Porting the misconception diagnoser into the hand-authored static feedback makes wrong answers teach *with AI off*, and keeps the model (when on) a phrasing layer over the same diagnosis.

The consensus counter-view, stated honestly and at full strength. Explanatory feedback ("why it's wrong," not a bare red X) beats verification feedback, and the obvious way to get it is to let the LLM generate it, tuned to what the learner actually did, the assignment even lists "explain a wrong answer in plain language, tuned to what the learner

actually did" as a prime AI candidate. And the model genuinely is good at this: warm, fluent, adaptive phrasing is exactly its strength, and the AI layer already paired it with a deterministic diagnosis to ground Koji's reveal.

The spiky case for deterministic feedback. This is a direct corollary of **SPOV 2 (the verification firewall)** and **SPOV 4 (AI-off-first)**: if the *diagnosis*, the identification of which misconception produced this wrong answer, is the valuable part, then it should not live on the AI path at all, because the AI path is exactly the one that can be off, slow, or wrong. The "why" of a Pythagorean mistake is *computable* from typed state, so the app ports the deterministic classifier (`diagnosis.ts` , added-the-legs, forgot-the-square-root, subtracted-legs, counted-one-square, picked-a-leg-not-the-hypotenuse, ...) out of the AI tools and into the hand-authored static **Feedback** , structured as **what you did → why it's wrong → the principle → a nudge to retry** (never the answer):

- The classifier becomes a **pure function** the static-feedback layer calls, no network, sub-100ms, living right next to the deterministic grading engine.
- Its output is **answer-free** by construction, so it can never leak the value: it teaches the *principle*, consistent with the answer-withholding firewall (SPOV 3).
- With AI **on**, Koji phrases warmly *over the identical diagnosis*, one source of truth for "why wrong," so the two paths can never disagree about the math.
- The result: the AI-**off** experience is now as *diagnostically* sharp as AI-on; the model adds tone and dialogue, not correctness.

The honest bound. Deterministic diagnosis only covers misconceptions we can characterize from typed state; a genuinely novel error falls back to a general "let's walk through it" nudge (the engine still owns correctness), and Koji, when on, can extemporize beyond the catalog, but never past the firewall.

DOK 3: Insights

Synthesized realizations that bridge the Knowledge Tree (DOK 2) and the SPOVs (DOK 4). Each carries an explicit SPOV connection.

Insight 1: For home learning, engagement is the binding constraint, so the modality that keeps the learner *in the loop* wins, even at some cost to per-rep encoding depth. Bloom's 2-sigma gains and the Freeman active-learning gains both

depend on the learner *continuing to participate* in a tight feedback loop. In a consumer setting the dominant failure mode isn't a shallow repetition; it's *abandonment*. Lowering the activation energy of the next interaction is therefore the highest-leverage design decision, and voice has the lowest activation energy of any input.

SPOV Connection: Primary support for **SPOV 1 (voice-first)**; also underpins **SPOV 3** (an earned answer keeps them in the loop rather than ending it).

Insight 2: Speaking your reasoning aloud is simultaneously the *lowest-friction input* and a *high-quality encoding act*, but only when the learner does the talking. The production effect means the spoken word benefits *the producer*. This collapses what looks like a trade-off ("convenient OR effective") into a synergy ("convenient AND effective"), conditional on a design that makes the *learner* reason aloud rather than passively receive Koji's explanation.

SPOV Connection: Core of **SPOV 1 (voice-first)**; the "learner must produce" condition is enforced by the answer-withholding persona and the talk-to-unlock reveal gate of **SPOV 3**.

Insight 3: An LLM that can be wrong about math is *disqualifying*, so reliability cannot be a prompt property: it must be an architectural one. Because the cost of a confidently-wrong math answer is categorical (not marginal), no amount of prompt tuning reaches an acceptable error rate. The only sound response is to move correctness *out of the model* into a deterministic checker the model cannot override. Khanmigo's independently-invented "math agent" is convergent evidence.

SPOV Connection: Direct foundation of **SPOV 2 (verification firewall)**; also enables **SPOV 4** by guaranteeing a correct fallback always exists.

Insight 4: The single most valuable behavior an AI tutor can exhibit is *restraint*: withholding the answer to protect retrieval practice. Retrieval practice is the best-evidenced mechanism in learning science, and learners are metacognitively blind to it: they will *ask* for the very thing that undermines their learning. A good tutor must therefore protect the learner from their own (rational-feeling) request for the answer.

SPOV Connection: Core of **SPOV 3 (effort-gated reveal)**; the restraint is only safe to enforce because **SPOV 4** guarantees a graceful, complete experience around it.

Insight 5: "AI-native" is a *leverage claim*, not a *dependency claim*. A product that breaks when the model is unavailable has taken on risk, not added capability. The defensible version of "AI-native" is an app that fully works without AI and is *multiplied* by it, which is also the only way the firewall and the restraint are cheap to guarantee (there's always a correct hand-authored thing to fall back to).

SPOV Connection: Core of **SPOV 4 (additive, not load-bearing)**; the precondition that makes **SPOV 2** and **SPOV 3** practical to ship.

Insight 6: Typed state is a better prompt than rendered text: what the model *knows* matters more than how you *ask*. The structured object that produced the screen is a richer, cheaper, less ambiguous context than the screen itself. Grounding in typed state shrinks prompts, cuts cost, reduces hallucination, and (because the model and the verifier reason over the same objects) makes the firewall coherent.

SPOV Connection: Core of **SPOV 5 (context engineering)**; the structured correct answer it carries is what makes **SPOV 2's** leak-checks and engine-computed reveals possible.

Insight 7: Mayer's modality boundary condition is a *design constraint*, not a *refutation of voice*. Printed words can beat spoken words for symbol-heavy, technical material, and math is symbol-heavy. The correct reading isn't "don't use voice"; it's "keep the *symbols* visual and make the *conversation* audio." Koji talks; the equation and triangle stay on screen. Voice and visuals are split along exactly the channel boundary Mayer's theory prescribes.

SPOV Connection: Strengthens and honestly bounds **SPOV 1 (voice-first)**; an application of the dual-channel basis from the Multimedia & Modality knowledge category.

Insight 8: Mastery should be measured by *durability*, not by a single correct moment. Performance during or just after study is a poor proxy for retention (Soderstrom & Bjork, 2015), and the spacing effect proves a fact you can produce today can vanish in a week. Fusing the mastery gate with a spaced review turns "mastery" from a screenshot into a time-series: you have mastered a skill when it *survives* a delay, not when you pass once.

SPOV Connection: Core of **SPOV 6 (durable mastery)**; depends on the scheduler of **SPOV 7** and the retrieval emphasis of **SPOV 8**; extends Bloom with Bjork's learning-vs-performance distinction.

Insight 9: For scheduling, the leverage is a *correct memory model and a clean outcome mapping*, not a per-user-trained one. On a seven-skill course the marginal accuracy of trained-vs-default FSRS is small; what matters is (a) using a model that actually fits a forgetting curve, so "when to review" is a prediction rather than a multiplier, and (b) mapping "couldn't produce it unaided" → lapse so misses resurface soonest. Defaults that are *right in shape* beat a hand-tuned multiplier with no memory model.

SPOV Connection: Core of **SPOV 7 (evidence-based scheduler)**; the honest defaults-vs-trained tradeoff mirrors how this document treats Bloom's 2-sigma as directional, not literal.

Insight 10: A review that feels easy is usually the wrong review. Recognition reviews feel productive and are metacognitively flattering; retrieval reviews feel harder and teach more. Because learners are *blind* to this gap (Roediger & Karpicke), the *system* must choose the harder, generative, cumulative review on their behalf, the same protective paternalism that justifies answer-withholding, now applied to review design.

SPOV Connection: Core of **SPOV 8 (retrieval over recognition)**; the scaled-up sibling of **SPOV 3 (effort-gated reveal)**; grounded in the retrieval / desirable-difficulty cluster.

Insight 11: If the diagnosis is the valuable part of feedback, it must not live on the AI path. The model's gift is *phrasing*, not knowing *why* an answer is wrong, and that "why" is computable from typed state. Moving the diagnoser into the deterministic, hand-authored layer makes the AI-off experience diagnostically equal to AI-on and gives the model, when on, a single source of truth to phrase over (so the two paths can never disagree about the math).

SPOV Connection: Core of **SPOV 9 (deterministic feedback)**; a direct corollary of **SPOV 2 (verification firewall)** and **SPOV 4 (additive / AI-off-first)**; the same instinct as **SPOV 5 (context engineering)**, reason over typed state, not rendered text.

Insight 12: A verification firewall should be scoped to where truth is taught, not reflexively applied to everything checkable. The same architecture can be absolute in one place and a calculated trust in another, and the dividing line is *stakes*, not checkability. Where the app presents an answer *as correct*, the hand-authored lessons, the model is never allowed to decide truth. Where content is merely *practiced*, supplemental generated problems, a strong reasoning model constrained by a careful prompt is trusted to author correctly, accepting a small, bounded, honestly-stated risk in exchange for the variety the deterministic verifier was quietly suppressing. The firewall is absolute where truth is *taught*, and a calculated trust where content is only *practiced*.

SPOV Connection: Scopes **SPOV 2 (verification firewall)** without weakening it for the core engine; leans on **SPOV 4 (additive / AI-off-first)** for the safety net beneath the bet and serves the variety **SPOV 8** depends on; argued in full in **Appendix B.7**.

Experts

Researched for accuracy of frameworks, contributions, and handles. Justin Sung is primary (the owner's anchor); the roster is kept at six by folding closely-related researchers (Bjork, MacLeod; Ericsson) into the entries they're most associated with.

1. Justin Sung (*primary*)

Who: Former medical doctor turned learning coach; co-founder of **iCanStudy**; widely-followed educator on the science of *how* to learn.

Focus / main views: Sung's central claim is that most students fail not at *retrieval* or *storage* but at **encoding**, and specifically at *higher-quality* encoding, i.e. building integrated, relational, **higher-order knowledge structures** rather than shallow ones (a constructivist framing tied to the upper tiers of **Bloom's taxonomy**). He argues retrieval practice (flashcards, testing) is genuinely useful but *easy to execute and over-relied upon*, while the harder, higher-leverage skill (elaborative, self-regulated encoding) is what students neglect. He teaches **cognitive load management**, **metacognition / self-regulated learning**, **SIR** (spaced interleaved retrieval), and the **GRINDE** mind-mapping framework (**G**rouped, **R**eflective/relational, **I**nterconnected, **N**on-verbal, **D**irectional,

Emphasized) as a tool for forcing high-order encoding. He is pointedly critical of passive techniques: rereading, highlighting, and leaning on flashcards/spaced repetition *without* good encoding underneath.

Why Follow: Sung is the owner's anchor expert and serves a deliberately *double* role in this BrainLift. He **supports** the active-processing design: Koji is built to make the learner *do the cognitive work* (reason aloud, attempt before reveal) rather than receive answers, which is exactly Sung's encoding-first prescription. And he **honestly complicates SPOV 1**: Sung is the expert most likely to warn that voice can decay into passive listening, the lowest-quality encoding state. Holding both at once is what makes the voice SPOV a real argument rather than a slogan: the product's answer-withholding persona and talk-to-unlock reveal gate are precisely the mechanisms that keep voice on the active-encoding side of Sung's line.

Where:

- iCanStudy: <https://icanstudy.com/>
- YouTube: <https://www.youtube.com/@JustinSung>
- Reference notes on his framework: <https://notes.alexandriathylane.com/Learning-Science--and--Pedagogy/iCanStudy---Justin-Sung/Justin-Sung-on-learning>

2. Benjamin Bloom

Who: Educational psychologist (1913–1999), University of Chicago; author of Bloom's taxonomy and originator of the **2-sigma problem**.

Focus / main views: Bloom (1984) reported that students tutored one-to-one with **mastery-learning** techniques performed about **two standard deviations** above conventionally-taught peers (the average tutored student outscoring 98% of the control class), with mastery learning alone worth about +1 sigma. He framed this not as a victory but as a *problem*: one-to-one tutoring is too costly to scale, so the challenge is to find group methods that approach its effect. He is also the namesake of **Bloom's taxonomy** of cognitive objectives (remember → understand → apply → analyze → evaluate → create).

Why Follow: The 2-sigma problem is the foundational business and pedagogical case for an AI tutor: software is the first plausible way to give *every* learner a responsive, mastery-oriented 1:1 tutor at near-zero marginal cost. His taxonomy is also the backbone of Sung's

"higher-order learning" language. Cited honestly: modern meta-analyses suggest the original 2-sigma figure is likely overstated (real tutoring effects often land nearer 0.3–0.7 SD), so this BrainLift treats "2 sigma" as a directional north star, not a literal promise.

Where:

- Bloom, B. S. (1984). *The 2 Sigma Problem*. Educational Researcher, 13(6): <https://gwern.net/doc/psychology/1984-bloom.pdf>
- Overview: https://en.wikipedia.org/wiki/Bloom%27s_2_sigma_problem

3. Richard E. Mayer

Who: Distinguished Professor of Psychology, UC Santa Barbara; originator of the **Cognitive Theory of Multimedia Learning (CTML)**.

Focus / main views: CTML rests on three assumptions: **dual channels** (separate visual/pictorial and auditory/verbal processing), **limited capacity** per channel, and **active processing**. From these Mayer derives design principles, including the **modality principle** (people learn more deeply from graphics + *spoken* words than graphics + on-screen text; median $d \approx 1.0$), the **redundancy principle** (graphics + narration beats graphics + narration + identical on-screen text; $d \approx 0.86$), and **coherence** (remove extraneous material; $d \approx 0.86$). He also documents *boundary conditions*, e.g. printed words can be preferable for technical terms/symbols, non-native speakers, or hearing-impaired learners.

Why Follow: Mayer supplies the cognitive-load basis for the voice SPOV: speaking while the eyes stay on the figure offloads the verbal channel and protects the visual channel from overload. Equally important, his boundary condition (text can win for symbol-heavy material) is the honest constraint that shapes the design: keep the *equation and triangle visual*, make the *conversation audio*. Mayer turns "voice feels nicer" into a falsifiable, channel-level claim.

Where:

- Modality principle (Cambridge Handbook): <https://www.cambridge.org/core/books/multimedia-learning/modality-principle/E5CD6E01CEA0B568CE260F66A3CD0D1F>
- Mayer, R. E. (2017). *Using multimedia for e-learning*. J. Computer Assisted Learning: <https://onlinelibrary.wiley.com/doi/10.1111/jcal.12197>

4. Henry L. Roediger III & Jeffrey D. Karpicke (*with Robert A. Bjork & Colin M. MacLeod*)

Who: Roediger (Washington University in St. Louis) and Karpicke (Purdue) are the leading modern researchers of the **testing/retrieval effect**; Robert A. Bjork (UCLA) coined **desirable difficulties**; Colin M. MacLeod (University of Waterloo) named and characterized the **production effect**.

Focus / main views: Roediger & Karpicke (2006) and Karpicke & Roediger (2008, *Science*) show that **retrieval practice**, effortfully producing an answer from memory, produces far better long-term retention than restudying, and that learners are **metacognitively blind** to this benefit. Bjork's **desirable difficulties** (1994) frames such effortful conditions as *good for retention even though they slow initial performance*. The **generation effect** (Slamecka & Graf, 1978) shows self-generated material is better remembered than read material. MacLeod, Gopie, Hourihan, Neary & Ozubko (2010) named the **production effect**: producing information *aloud* beats silent reading via added encoding *distinctiveness*.

Why Follow: This cluster is the scientific backbone of two SPOVs at once. The testing/generation/desirable-difficulty work is the case for **answer-withholding** (SPOV 3): the answer must be *retrieved*, not received. The production effect is a load-bearing pillar of **voice-first** (SPOV 1): when the *learner* speaks their reasoning, voice is a strong encoding act. Cited honestly: the production effect is strongest in *mixed-list* designs and improves *retention/distinctiveness*, not deep comprehension, which is exactly why this BrainLift positions voice as an engagement-and-retention lever, with comprehension carried by active reasoning + the engine + visuals.

Where:

- Karpicke & Roediger (2008), *The Critical Importance of Retrieval for Learning*, *Science*: <https://www.science.org/doi/10.1126/science.1152408>
- MacLeod & Bodner (2017), *The Production Effect in Memory*, *Current Directions in Psychological Science*: <https://journals.sagepub.com/doi/10.1177/0963721417691356>
- Roediger & Karpicke (2006), testing-effect article (PDF): http://psychnet.wustl.edu/memory/wp-content/uploads/2018/04/Roediger-Karpicke-2006_PPS.pdf

5. Sal Khan / Khanmigo (Khan Academy)

Who: Sal Khan, founder of Khan Academy; **Khanmigo** is its GPT-4-based AI tutor (piloted March 2023).

Focus / main views: Khanmigo is explicitly designed to **guide rather than give answers**: *"unlike other AI tools such as ChatGPT, Khanmigo doesn't just give answers... it guides learners to find the answer themselves,"* using Socratic prompting. To handle math reliably, Khan Academy added a behind-the-scenes **"math agent"** that verifies calculations and checks expressions against *verified content*, generated textual representations of graphics so the model can "see" them, and tracks **guardrail metrics** including math error rate and *"giving the answer away before a student submitted."*

Why Follow: Khanmigo is the real-world precedent for Koji on *both* contested SPOVs. Its answer-withholding Socratic stance validates **SPOV 3 (restraint)** at production scale. Its math-agent-over-verified-content architecture is independent convergent evidence for **SPOV 2 (verification firewall)**: the same conclusion that *generative models must be demoted below a deterministic checker for math*. And Khan Academy's candid admission that genAI "was not designed to handle mathematical reasoning" is the clearest statement of the hallucination risk the firewall exists to neutralize.

Where:

- Khanmigo: <https://www.khanmigo.ai/>
- Khan Academy (2024), *Khanmigo Math Computation and Tutoring Updates*: <https://blog.khanacademy.org/khanmigo-math-computation-and-tutoring-updates/>
- *How Khan Academy Is Building a Better AI Tutor*: <https://blog.khanacademy.org/how-khan-academy-is-building-a-better-ai-tutor-our-most-recent-learnings/>

6. Carl Wieman (with Anders Ericsson, deliberate practice)

Who: Nobel-laureate physicist (Stanford) turned science-education researcher; a leading champion of **active learning**. Anders Ericsson (1947–2020) was the originator of **deliberate practice**.

Focus / main views: Wieman argues that learning is doing: students must actively grapple with problems, not passively receive lectures. The landmark Freeman et al. (2014) PNAS meta-analysis (225 STEM studies), which Wieman championed, found active learning raised exam scores by ~0.47 SD and that students in traditional lecture courses were **1.5×**

more likely to fail (33.8% vs 21.8%). Ericsson's **deliberate practice** adds that expertise comes from *effortful, feedback-rich, targeted* practice at the edge of current ability, not mere repetition.

Why Follow: Wieman is the empirical case that *learning-by-doing beats passive reception*: the foundation of AlphaBrilliant's entire "learn-by-doing course" premise and a direct support for **SPOV 1** (voice that keeps learners *actively participating*) and **SPOV 3** (productive struggle over spoon-fed answers). Ericsson's deliberate-practice lens motivates **Infinite Practice**: targeted problems at adaptive difficulty with immediate feedback (the problems now model-authored under prompt constraints, a scoped exception to the firewall; see SPOV 2 and Appendix B.7).

Where:

- Freeman et al. (2014), *Active learning increases student performance in STEM*, PNAS: <https://www.pnas.org/doi/10.1073/pnas.1319030111>
- Wieman, C. (2014), *Large-scale comparison of science teaching methods sends clear message*, PNAS commentary: <https://www.pnas.org/doi/10.1073/pnas.1407304111>

DOK 2: Knowledge Tree

Organized broad → specific: each category lists **DOK 1: Facts** (concrete, cited) then a **DOK 2: Summary & Analysis** that ties the facts to the product and to specific SPOVs.

Category 1: The 2-Sigma Problem & Mastery Learning

1.1 One-to-one tutoring + mastery learning

Source: Bloom, B. S. (1984). *The 2 Sigma Problem: The Search for Methods of Group Instruction as Effective as One-to-One Tutoring*. Educational Researcher, 13(6), 4–16.
<https://gwern.net/doc/psychology/1984-bloom.pdf>

DOK 1: Facts

- The average one-to-one-tutored student (with mastery learning) scored **~2 SD** above the conventional-class average, outperforming **98%** of the control class.
- Mastery learning *alone* (in a 30-student class) produced **~+1 SD** (above ~84% of the control class).

- Time-on-task rose from ~65% (conventional) to ~75% (mastery) to ~90%+ (tutoring); attitudes/interest were most positive under tutoring.
- Bloom framed the result as a *scalability problem*: tutoring is "too costly for most societies to bear on a large scale."
- Later meta-analyses suggest the 2-sigma figure is likely overstated; real tutoring effects often land closer to 0.3–0.7 SD.

DOK 2: Summary & Analysis Bloom is *why an AI tutor is worth building at all*: he quantified the prize (a responsive, mastery-oriented 1:1 tutor) and named the obstacle (cost/scale) that software is uniquely positioned to dissolve. Note the mechanism behind his numbers: *time-on-task* climbs steeply with tutoring. That is the empirical seed of **Insight 1** and **SPOV 1**: the tutoring advantage is, in large part, an *engagement* advantage, which is exactly what a low-friction voice companion is designed to maximize at home. It also motivates **SPOV 3**: tutoring works through tight feedback and correction loops, not answer-dispensing. Cited honestly (the overstatement caveat), it functions here as a directional north star, not a promised effect size.

1.2 Mastery learning's corrective loop (formative → corrective → re-test)

Source: Bloom, B. S. (1968). *Learning for Mastery*. Evaluation Comment, 1(2); and Bloom (1984, above). Overview: https://en.wikipedia.org/wiki/Mastery_learning

DOK 1: Facts

- Bloom's **mastery learning** cycle: teach a unit → give an ungraded **formative** assessment that *diagnoses* specific gaps → deliver targeted **corrective** instruction on those gaps → **re-test** to confirm mastery before advancing.
- The corrective step is meant to be *different* from the original teaching (re-approach the gap), not a literal repeat.
- The high mastery bar (often ~80–90%) plus this feedback-corrective cycle is what Bloom credits for mastery learning's ~+1 SD (Bloom, 1984).
- Honest: realized effect sizes vary widely with implementation fidelity; the loop costs time and only pays off when the corrective work genuinely targets the diagnosed error.

DOK 2: Summary & Analysis This subsection is the spine that ties the three retention-scheduling SPOVs into one mechanism rather than three bolt-ons. The app implements Bloom's loop almost literally: the **formative** assessment is the deterministic misconception diagnosis (the same classifier ported in **SPOV 9**), the **corrective** step is targeted Infinite-

Practice on the diagnosed skill, and the **re-test** is a *spaced* FSRs review (**SPOV 7**), which is also the durability check that defines mastery in **SPOV 6**. The per-skill **mastery meter** is Bloom's "clear mastery signal." Read this way, spacing, durable mastery, and sharpened feedback are not separate features; they are the formative→corrective→re-test loop with FSRs as the re-test scheduler and the deterministic diagnoser as the formative assessment.

Category 2: Encoding & Higher-Order Learning (Sung)

2.1 Encoding vs. retrieval; higher-order knowledge structures; GRINDE

Source: Justin Sung / iCanStudy, interview notes and framework summaries.

<https://notes.alexandriathylane.com/Learning-Science--and--Pedagogy/iCanStudy---Justin-Sung/Justin-Sung-on-learning> · <https://www.youtube.com/@JustinSung>

DOK 1: Facts

- Sung distinguishes **encoding** (forming knowledge structures) from **storage/retrieval** (accessing them) and argues most students fail at *higher-quality encoding*.
- "Higher-quality encoding" = higher-order thinking that builds **integrated, relational knowledge structures**, correlated with more durable, meaningful retention (a constructivist framing).
- He teaches **SIR** (spaced interleaved retrieval) and warns that retrieval practice is *easy to execute and over-relied upon* relative to harder encoding skills (elaboration, self-explanation, mind-mapping).
- **GRINDE** mind-mapping = **G**rouped, **R**eflective, **I**nterconnected, **N**on-verbal, **D**irectional, **E**mphasized: a protocol for forcing high-order encoding.
- He is critical of passive techniques: rereading, highlighting, and flashcards/spaced-rep *without* good encoding.

DOK 2: Summary & Analysis Sung is the lens that keeps the voice bet honest. His encoding-first view *demand*s that the learner do active cognitive work, and the product obliges: Koji withholds answers and pushes the learner to reason, and the reveal is unlocked by *engaging* (ideally *talking through*) the problem. But Sung is also the strongest internal critic of **SPOV 1**: he'd flag that listening to Koji explain is exactly the passive, low-quality encoding he warns against. The resolution baked into the design (learner produces, Koji nudges) is what moves voice from passive reception to active encoding, satisfying Sung

rather than contradicting him. His higher-order/Bloom framing also informs why generation is restricted to *problems within the proven schema*: the app doesn't ask the model to invent pedagogy.

Category 3: Retrieval, Generation & Production Effects (the "speak aloud" evidence)

3.1 The testing / retrieval-practice effect

Source: Karpicke, J. D., & Roediger, H. L. (2008). *The Critical Importance of Retrieval for Learning*. *Science*, 319(5865), 966–968.

<https://www.science.org/doi/10.1126/science.1152408>

DOK 1: Facts

- Repeated **testing** (retrieval) produced a large positive effect on delayed recall; repeated **studying after learning** had essentially *no* added effect.
- Students' **predictions of their own performance were uncorrelated** with actual retention: a metacognitive blind spot.
- The effect is robust across materials and designs (Roediger & Karpicke, 2006).

3.2 The generation effect & desirable difficulties

Source: Slamecka & Graf (1978), the generation effect; Bjork, R. A. (1994), *desirable difficulties* (see Soderstrom & Bjork, 2015). Overview via Roediger & Karpicke (2011), *Intricacies of Spaced Retrieval*. <http://psychnet.wustl.edu/memory/wp-content/uploads/2018/04/Roediger-Karpicke-2011.pdf>

DOK 1: Facts

- **Generation effect:** self-generated items are remembered better than read items.
- **Desirable difficulties:** conditions that *slow* acquisition (testing, generation, varied/spaced practice) often *improve* long-term retention and transfer.
- Boundary: desirable difficulties can become *undesirable* when working-memory load (element interactivity) is too high (cognitive-load moderation).

3.3 The production effect

Source: MacLeod, C. M., & Bodner, G. E. (2017). *The Production Effect in Memory*. *Current Directions in Psychological Science*.

<https://journals.sagepub.com/doi/10.1177/0963721417691356> (effect named by MacLeod, Gopie, Hourihan, Neary & Ozubko, 2010).

DOK 1: Facts

- Producing material **aloud** yields substantially better memory than silent reading; the mechanism is encoding **distinctiveness**.
- Speaking gives the largest benefit, but writing, typing, and even mouthing also help.
- The effect is **strongest in mixed-list designs** (some items produced, some not); producing *everything* can shrink it.
- It improves **retention/recognition**, but **not** deep comprehension or inference.

DOK 2: Summary & Analysis This is the category that makes **SPOV 1** more than a convenience argument and **SPOV 3** more than a UX preference. The testing/generation/desirable-difficulty research is the hard case for **answer-withholding**: the learner must *retrieve and generate*, not receive, and they will be *blind to why that's good*, which is exactly why restraint must be enforced in code (the reveal tool's effort gate) rather than left to the learner's judgment. The production effect is the pillar under voice-first: when the *learner speaks their reasoning*, voice becomes a distinctiveness-boosting encoding act, turning the "convenient vs. effective" trade-off into a synergy (**Insight 2**). The honesty is built into the citations: production helps retention, not comprehension, and shines in *mixed* production, so the design treats voice as one channel among several (visuals + verified engine carry comprehension) rather than a cure-all, and keeps both spoken and typed turns available.

Category 4: Multimedia & Modality (the cognitive-load basis for voice)

4.1 The Cognitive Theory of Multimedia Learning; the modality & redundancy principles

Source: Mayer, R. E. (2005/2009). *Modality Principle*, in The Cambridge Handbook of Multimedia Learning; Mayer (2017), *Using multimedia for e-learning*, JCAL.

<https://www.cambridge.org/core/books/multimedia-learning/modality-principle/E5CD6E01CEA0B568CE260F66A3CD0D1F> ·
<https://onlinelibrary.wiley.com/doi/10.1111/jcal.12197>

DOK 1: Facts

- CTML assumes **dual channels** (visual + auditory), **limited capacity** per channel, and **active processing**.

- **Modality principle:** deeper learning from graphics + *spoken* words than graphics + on-screen text (median $d \approx 1.0$; 17/17 tests in one review): audio offloads words to the verbal channel and frees the visual channel for the figure.
- **Redundancy principle:** graphics + narration beats graphics + narration + identical on-screen text ($d \approx 0.86$).
- **Boundary conditions:** printed words may be preferable for **technical terms/symbols**, non-native speakers, or hearing-impaired learners; modality helps most when material is complex and fast-paced.

DOK 2: Summary & Analysis Mayer converts the voice bet from aesthetic preference into a channel-level, falsifiable claim, and supplies its honest constraint. The supportive read (**SPOV 1**): Koji *speaks* while the triangle/equation stay on screen, so the learner's eyes process the figure while the ears process the words: textbook modality-principle design, and the reason voice + visuals beats text-on-text here. The constraining read (**Insight 7**): because math is symbol-heavy, the boundary condition warns against trying to convey the *equation itself* by voice. The design respects this exactly: symbols remain *visual* (on-screen KaTeX), conversation is *audio*. This is also why the build skips AI-generated visuals and keeps the hand-built visual templates: the visual channel is too important to hand to an unverified generator.

Category 5: Active Learning & Deliberate Practice

5.1 Active learning vs. lecture (large-scale meta-analysis)

Source: Freeman, S., et al. (2014). *Active learning increases student performance in science, engineering, and mathematics*. PNAS, 111(23), 8410–8415.

<https://www.pnas.org/doi/10.1073/pnas.1319030111> (commentary: Wieman, 2014).

DOK 1: Facts

- Meta-analysis of **225 STEM studies**: active learning raised exam/concept-inventory scores by **~0.47 SD** (~6% on exams).
- Failure rates were **33.8% under lecture vs 21.8% under active learning**: lecture students **1.5× more likely to fail** (a 55% relative increase).
- Effects held across disciplines and class sizes; largest in small classes; robust to publication-bias checks.
- Ericsson's **deliberate practice**: expertise comes from effortful, feedback-rich practice targeted at the edge of current ability.

DOK 2: Summary & Analysis This is the empirical floor under AlphaBrilliant's whole premise (a *learn-by-doing* course, not a video lecture) and a direct support for both **SPOV 1** and **SPOV 3**. "Active" is not décor; it is the variable that moves failure rates by half. Voice-first matters here because the cheapest way to *keep* a learner active at home is to remove input friction (**Insight 1**). Deliberate practice grounds **Infinite Practice**: not random drills but *adaptive-difficulty* problems with immediate feedback: difficulty derived from the learner's own attempt history (first-try rate ≥ 0.8 and low attempts $\rightarrow$ harder; struggling $\rightarrow$ easier), with each problem now *model-authored* under prompt constraints rather than deterministically re-derived before display, a deliberate, scoped exception to the firewall (SPOV 2, Appendix B.7).

Category 6: AI Tutors in Practice & Their Failure Modes

6.1 Khanmigo: Socratic restraint + a math-verification agent

Source: Khan Academy (2024). *Khanmigo Math Computation and Tutoring Updates* and *How Khan Academy Is Building a Better AI Tutor*.

<https://blog.khanacademy.org/khanmigo-math-computation-and-tutoring-updates/> · <https://blog.khanacademy.org/how-khan-academy-is-building-a-better-ai-tutor-our-most-recent-learnings/>

DOK 1: Facts

- Khanmigo is designed **not to give direct answers**; it uses Socratic questioning to guide students to the answer themselves.
- Generative AI "was not designed to handle mathematical reasoning... it predicts the most probable numbers... not always the correct answer."
- Khan Academy added a behind-the-scenes **"math agent"** that verifies calculations and checks expressions against **verified content**; it also generates **textual descriptions of graphics** so the model can "see" them.
- Guardrail metrics tracked include **math error rate** and **"giving the answer away before a student submitted"**; prompts require evaluation harnesses because identical prompts yield different outputs and long prompts get partially ignored.

DOK 2: Summary & Analysis Khanmigo is the strongest real-world corroboration in this BrainLift, and it lands on *three* SPOVs at once. (1) Its answer-withholding Socratic design is independent validation of **SPOV 3 (restraint)**, and it operationalizes the very gate Koji enforces ("don't give the answer away before they've attempted"). (2) Its math-agent-over-

verified-content architecture is convergent evidence for **SPOV 2 (verification firewall)**: a leading team independently concluded that *the model must be demoted below a deterministic checker for math*. (3) Khan's note that prompts are non-deterministic and partially-ignored validates **SPOV 2** further: you cannot prompt your way to reliability, so for everything it *teaches as true* AlphaBrilliant does the same thing structurally, a deterministic grading engine that owns correctness, engine-computed reveals, and server/client leak firewalls. (Supplemental generated practice is the one scoped exception, where correctness is trusted to a strong reasoning model rather than re-derived; see SPOV 2 and Appendix B.7.) Khan's practice of grounding in *typed/verified content* (and converting graphics to text) is the same instinct as **SPOV 5 (context engineering)**: ground in structured truth, not rendered pixels.

6.2 Hallucination as a categorical risk for teaching

Source: Synthesis of Khan Academy (2024, above) and the structural nature of LLMs.

DOK 1: Facts

- LLMs generate the *most probable* token sequence, which for arithmetic need not be the correct value.
- For a tutor, a confidently-wrong answer mis-teaches with authority: an asymmetric, categorical cost, not a marginal one.

DOK 2: Summary & Analysis This fact is the entire reason **SPOV 2** and **SPOV 4** exist. Because the downside is categorical, the response can't be "reduce the error rate a bit with prompting"; it must be "make the model unable to determine truth" (firewall) and "guarantee a correct hand-authored fallback always exists" (additive/AI-off-first). Together they convert an unreliable component into a safe one: the model phrases; the engine decides; and if the model is absent or wrong, the deterministic experience is right there underneath.

Category 7: Spacing & Scheduling (the science of *when* to review)

7.1 The spacing effect

Source: Cepeda, N. J., Pashler, H., Vul, E., Wixted, J. T., & Rohrer, D. (2006). *Distributed practice in verbal recall tasks: A review and quantitative synthesis*. Psychological Bulletin, 132(3), 354–380. <https://doi.org/10.1037/0033-2909.132.3.354> · temporal scaling: Cepeda et al. (2008), Psychological Science.

DOK 1: Facts

- **Distributed (spaced) practice** reliably yields better long-term retention than **massed** practice (cramming): the **spacing effect**, one of the oldest and most robust results in memory research (Ebbinghaus, 1885).
- Cepeda et al. (2006) quantitatively synthesized the distributed-practice literature (hundreds of experiments) and found a consistent, sizable spacing advantage across materials and designs.
- The *optimal* review gap **scales with the retention interval**: the longer you need to remember, the wider the best spacing (Cepeda et al., 2008, the "temporal ridgeline"); there is no single magic interval.
- **Reviewing too soon is close to wasted.** A successful retrieval strengthens long-term memory roughly in proportion to how *effortful* it was: retrieving an item while it is still fresh (recall still high) adds little, whereas retrieving it after some forgetting adds much more (the *retrieval-effort hypothesis*, Pyc & Rawson, 2009; consistent with Bjork & Bjork's storage-strength vs retrieval-strength account, 1992). In spacing terms, a gap shorter than the optimum sits on the rising side of Cepeda's temporal ridgeline, below peak retention, so a review pulled forward is a low-effort, low-yield repetition.
- Honest boundaries: most of this evidence is verbal-recall/lab; the optimal-gap function is approximate and noisy; and "expanding intervals beat equal intervals" is **contested**, equal spacing often does just as well.

DOK 2: Summary & Analysis This is the empirical floor under **SPOV 7** and half of **SPOV 6**. Spacing is *why a scheduler must exist at all* and *why an immediate mastery pass is insufficient*: the gap between reviews is exactly where forgetting happens, and therefore where durable mastery is proven or lost. The honesty cuts in a useful direction here: because the optimal-interval science is noisy and the expanding-vs-fixed debate is unsettled, we explicitly *do not* hand-pick "magic" intervals. We delegate interval choice to a memory model (FSRS, §7.2) whose single job is to estimate when recall decays to the target, and we keep the claim modest: spacing helps; the exact schedule is an *estimate*, not a law. This research also drives a concrete product decision (Appendix B.3): we removed the "review early" button, because a review pulled forward before recall has decayed is exactly the low-effort, low-benefit retrieval the spacing and retrieval-effort work warns against. The scheduler's whole value is choosing the gap; an early-review affordance would simply hand that decision back to the learner's (well-meaning but miscalibrated) impulse to study sooner.

7.2 FSRS-6: a benchmarked, open-source memory model (real-world precedent / tool)

Source: Open-source. Anki: <https://github.com/ankitects/anki> · FSRS-rs reference implementation: <https://github.com/open-spaced-repetition/fsrs-rs> · the algorithm spec: <https://github.com/open-spaced-repetition/fsrs4anki/wiki/The-Algorithm> · benchmark: <https://github.com/open-spaced-repetition/srs-benchmark>

Following this document's treatment of Khanmigo, FSRS/Anki enters as a real-world *tool and precedent*, **not** a seventh expert, the roster is deliberately capped at six.

DOK 1: Facts

- **FSRS** (Free Spaced Repetition Scheduler) is Anki's modern, built-in scheduler, the default for new profiles on recent installs, supplanting legacy SM-2; **FSRS-6** is the current version (21 parameters).
- It uses the **DSR memory model**, **Difficulty** (1–10; how hard the item is for this learner), **Stability** (days until recall probability falls to 90%), **Retrievability** (current probability of recall), and fits a **power-law forgetting curve** (a better fit than the older exponential model).
- It schedules each item to resurface when predicted **retrievability** decays to a **target retention** (default **0.90**); a failure ("**Again**") is a **lapse** that collapses stability, while Hard/Good/Easy are passes.
- Default weights are trained on a large public corpus (on the order of **700M reviews from ~10,000+ Anki users**). On the public `srs-benchmark` (~9,999 collections, ~350M reviews), FSRS has **lower log-loss than SM-2 for ~99.5–99.6%** of collections, *even using defaults*. Per-user optimization trims error further.
- **Legacy SM-2** (SuperMemo / old Anki) is a hand-tuned **ease-factor × interval-multiplier** with **no memory model and no target retention**, it reacts to grades but never *predicts* recall.
- Honest caveats: SM-2 wasn't designed to output probabilities (the benchmark adapts it to compare), and the headline "**20–30% fewer reviews**" comes from **simulation**, not a controlled classroom trial.

DOK 2: Summary & Analysis FSRS is the real-world precedent that makes **SPOV 7** concrete, the way Khanmigo grounds SPOV 2/3. Two properties earn it a place over the simpler SM-2/Leitner box. First, it has an actual **forgetting-curve model**, so "when to review" is a *prediction* (recall has decayed to target) rather than an opaque multiplier.

Second, its grade vocabulary maps cleanly onto our **deterministic outcomes**: first-try correct = pass, **wrong or assisted = lapse**, which is exactly the "resurface wrong items sooner" behavior the assignment asks for. We port FSRs-6 with default weights (SPOV 7's honest tradeoff), persist D/S/R **per skill** (not per flashcard) under `users/{uid}/...`, and drive every update through the one outcome chokepoint (`recordStep`). The benchmark's own caveats, SM-2's probability adaptation, and a simulation-based efficiency figure, are precisely why this BrainLift states "FSRS beats SM-2" as a strong *directional* result, the same stance it takes toward Bloom's 2-sigma.

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Appendix A: How the Engine and AI Layer Were Built

The original build-retrospective BrainLift, preserved. Where the body of this document is about the *thinking*, this appendix is about the *making*: tools, workflow, prompting strategies, the shipped/skipped/deferred ledger, and the AI-vs-human code split.

A.1 Tools & workflow

- **Cursor** as the editor/agent host, driving a **10-agent orchestrated swarm** with roles: ai-foundation, backend-functions, tools, tutor-text, voice, generation, reveal-pedagogy, ui-animation, code-reviewer, and qa-swarm-lead.
- **Disjoint ownership + git worktrees**. Each agent owned one slice and worked in a sibling worktree, on a feature branch, merged by PR into stacked integration branches (problem generation, then the Koji tutor, then the AI features). Cross-agent contracts were pinned in code so parallel work composed without collisions.

- **Ralph-style looping.** A standing task brief defines a binary Done-bar (both pillars, AI-never-wrong, AI-off-safe, green type-check / lint / build, UI bar); the loop iterates until every box is true rather than stopping at "looks done."
- **Multi-pass review cadence.** Code review re-reviews after fixes until explicit APPROVE; **Emil** design/animation runs ≥ 3 passes; the **impeccable-swarm** UI/UX QA runs partition→inspect→fix→re-verify loops; **ponytail** complexity passes (a per-diff review and a pre-promotion audit) cut over-engineering. The git history shows this directly (e.g. commits for backend review fixes, ponytail safe cuts, and Emil UI passes).

A.2 Prompting strategies that worked

1. **Grounded-state tool briefs.** Every agent was told to build prompts from *typed state* serialized as compact JSON, never scraped DOM/KaTeX. This kept prompts small, cheap, and correct. (*This is SPOV 5 in practice.*)
2. **"Verify with the engine; the model only phrases."** Ground truth is the deterministic grading engine plus an independent math check. Generated problems originally had to pass a grading **round-trip** before display (later removed for the generated-practice path, which is now model-authored, see Appendix B.7); reveals are engine-computed; the model is demoted to a phrasing layer. (*SPOV 2.*)
3. **Disjoint-ownership parallel agents with explicit contracts.** Rather than one agent doing everything, each owned a file set behind a typed interface, so foundation/backend/voice/generation could be built simultaneously and merged cleanly.
4. **Don't trust the prompt: add a code backstop.** Hint prompts say "DO NOT state the value," *and* a server-side leak firewall re-checks the output and swaps in a static hint if the answer slips through. Defense in depth, not vibes.
5. **AI-off-first / additive.** Build the fallback path first: every client wrapper early-returns a safe empty result when the AI feature flag is off, so no AI path can break the app. (*SPOV 4.*)

A.3 Decisions: shipped / skipped / deferred (engine and AI layer)

Decision	Status	Why
Koji grounded tutor: progressive text hints, deterministic-diagnosis explanations, realtime voice , app tool-control	✅ Shipped	The core "AI-native" ask: a tutor that <i>sees</i> lesson state and <i>operates</i> the app with the learner.
Effort-gated, personalized reveal	✅ Shipped	Protects the learn-by-doing loop: answer is earned (genuine attempt + Koji engagement), engine-computed, marked assisted.
Verified adaptive generation / "Infinite Practice"	✅ Shipped	Course never runs dry; every problem passes the verification firewall.
Voice-first Koji panel (mascot + mic + unified voice/text thread; typing rides the same realtime session)	✅ Shipped	Voice is the primary modality; text shares one thread.
Open web chatbot	❌ Skipped	Koji is grounded and tool-scoped, not a free-form chat box.
AI-generated visuals / whole lessons / feedback-rewrite	❌ Skipped	Use the safe hand-built visual templates and hand-written feedback (the latter is the AI-off fallback); generate problems <i>within</i> the proven schema, not new pedagogy.
Spaced repetition / mastery scheduling / full adaptive sequencing	➡ Deferred → learning-science layer	That's learning-science scheduling. The AI layer adapts only <i>difficulty</i> inside generation.

Update: the deferred row above is now **delivered**: spaced repetition, durable mastery scheduling, and mastery/SR-driven sequencing all shipped; see the learning-science ledger in **Appendix B.3**.

Update: the "*Verified adaptive generation*" row above describes the AI layer as *originally built*, when every generated problem was round-trip-verified by the engine. In the learning-science layer that round-trip was **deliberately removed** for the generated-practice path: Infinite Practice is now *model-authored* and prompt-constrained, with correctness trusted to the model rather than re-derived (the firewall stays absolute for the core teaching engine). See **SPOV 2** and **Appendix B.7**.

A.4 Code analysis: AI-generated vs hand-written

Honestly: **this build is ~entirely AI-generated and AI-orchestrated**. Effectively all shipped code (the AI layer, the backend functions, the API, the practice components, and the Koji UI) was written by the agent swarm. The human contribution was *direction and judgment*, not code: the PRD and locked decisions, the headline voice-first thesis, the agent role definitions and prompts, the architectural guardrails (additive flag, verification firewall, secrets server-side), and the review/merge gating. So the split is roughly **~100% AI-written code / 100% human-decided constraints**: the leverage was in framing the problem so the swarm couldn't ship something wrong.

A.5 Key learnings (the seeds of the SPOVs above)

- **A verification firewall beats prompt engineering.** Reliability came from code that *checks* the model (round-trip grading, leak detection), not from a cleverer prompt. → **SPOV 2.**
 - **Ground in typed state, not screen text.** Feeding the model the structured lesson state and the learner's typed answer made prompts smaller, cheaper, and far more accurate than scraping the rendered page. → **SPOV 5.**
 - **Build AI-off first; AI is the upgrade, not the product.** Because the flag-gated app is byte-for-byte the deterministic engine with AI off, the AI is never load-bearing and there's always a graceful fallback. → **SPOV 4.**
 - **Restraint is a feature.** The hardest pedagogy decision was making the tutor *refuse* the answer until it's earned. → **SPOV 3.**
 - **Voice is the main use case, designed to stay active.** Voice wins on engagement and (when the learner talks) on encoding, provided the design keeps the learner producing rather than passively listening. → **SPOV 1.**
 - **The swarm's bottleneck is review, not authoring.** Parallel agents produce code fast; the multi-pass review/ponytail cadence is what keeps quality and complexity in check, and counters the model's instinct to over-build.
-

Appendix B: How the Learning-Science Layer Was Built

Mirrors Appendix A for the learning-science layer. Where the body of this document is about the *thinking* behind retention, this appendix is about the *making*: the tools and workflow, the prompting strategies that worked, the shipped/skipped/deferred ledger, the AI-vs-human code split, and the key learnings (the seeds of SPOVs 6–9). The defining constraint carried throughout: everything here is **additive and AI-off-safe**, it rides existing primitives and adds no AI dependency.

B.1 Tools & workflow

- **Same host, isolated worktree.** Cursor as the agent host; the learning-science work lived in its own sibling worktree on a dedicated feature branch, never touching the frozen engine-and-AI snapshot in `main/`. The non-negotiable bar from `RALPH_TASK.md` carried over verbatim: additive, AI-off-safe, green `tsc / eslint / build`, UI bar held.
- **Two specialized agents ran first, before any feature code:**
 1. A **read-only codebase-mapping agent** traced the seams the learning-science layer had to attach to, the single outcome chokepoint (`recordStep` in `src/lib/learner.tsx`), the deterministic engine (`src/content/engine.ts`), the content types (`ProblemStep / Feedback` in `src/content/types.ts`), and the existing misconception diagnosis (`src/lib/ai/tools/diagnosis.ts`). This forced the new layer to ride real primitives instead of forking a parallel progress store.
 2. An **Anki / FSRS research agent** cloned `ankitects/anki` and `open-spaced-repetition/fsrs-rs`, read the FSRS-6 algorithm spec, extracted the default weights and the DSR + power-curve formulas, and produced a faithful **TypeScript port validated by unit tests**, the port's outputs checked against reference values from the Rust/wiki implementation so a plausible-but-wrong scheduler couldn't slip through.
- **Same multi-pass review cadence as the engine and AI layer.** Code review re-reviewed to an explicit APPROVE; **Emil** design passes on the new mastery meters and Reviews-due hub; the **impeccable-swarm** UI/UX QA loop; and **ponytail** complexity passes to keep an additive layer from bloating (the FSRS port was the one place we *accepted* added complexity, justified by the unit-test bar).
- **Done-bar, extended.** The same bar as before, plus one new line: *the FSRS port matches reference outputs in unit tests, and every new state path is exercised with the AI flag off*.

B.2 Prompting strategies that worked

1. **"Attach to the chokepoint; don't fork state."** Every SR/mastery update had to flow through the existing `recordStep` outcome path and nest under `users/{uid}/...`, no second writer. This kept the AI-on and AI-off paths writing one truth, and kept the layer additive. (*SPOV 4, SPOV 7.*)
2. **"Port the algorithm, then prove it with tests, *before* any UI."** The FSRs agent was told to land a unit-test-validated TS port of FSRs-6 (default weights, DSR model, power curve, target retention 0.9) and only *then* wire scheduling, correctness settled before product. (*SPOV 7.*)
3. **"Map the outcome→grade boundary explicitly, once."** Pin the rule that **wrong/assisted = lapse** and **first-try-correct = pass**, encoded at the chokepoint, so "resurface what they got wrong, sooner" is a property of the data path, not of any screen. (*SPOV 7.*)
4. **"Port the diagnoser across the AI line."** Move `diagnosis.ts`'s classifier into the deterministic static-feedback layer (*what you did* → *why* → *principle* → *nudge*, never the answer), with AI-on phrasing over the *same* diagnosis. (*SPOV 9.*)
5. **"Mastery = survives a spaced review."** Encode durable mastery and the level gate as the spec up front, so agents couldn't regress to a one-pass mastery flag or per-lesson locking. (*SPOV 6.*)

B.3 Decisions: shipped / skipped / deferred (learning-science layer)

Decision	Status	Why
FSRS-6 spaced repetition: per-skill D/S/R, target retention 0.9, wrong/assisted = lapse ; "Reviews due (N)" hub + due-soon forecast + one-tap session; captures Infinite-Practice misses	 Shipped	Resurfaces each skill as predicted recall decays, and misses soonest, the assignment's headline SR ask, from a real memory model, all AI-off-safe.

Decision	Status	Why
No "review early" affordance	 Shipped	A review only helps once recall has decayed enough to make retrieval effortful; pulling one forward while the item is still fresh spends the rep for little gain (Category 7.1: retrieval-effort + spacing). The home card shows the due-soon forecast, not an early-review button, so the FSRS-computed gap is respected. Adapting Anki's scheduler means more than porting it: it means removing the affordance that would let a learner defeat it.
Durable, gated mastery: per-skill mastery meter , "mastery = survives a spaced review," Bloom corrective loop, mastery-driven next step, level + level-review lock until mastered	 Shipped	A real mastery gate with a clear signal, fused with spacing so it certifies <i>learning</i> , not a moment of performance.
Retrieval-first reviews: cumulative (interleaved) level review, opening recall warm-up, generative + scaffold-dropped item selection	 Shipped	Recall over recognition (a desirable difficulty); this is also where interleaving lives, folded in, not a headline.
Ported static misconception feedback: deterministic diagnoser in the hand-authored Feedback (<i>what you did</i> → <i>why</i> → <i>principle</i> → <i>nudge</i>)	 Shipped	Wrong answers teach with AI off ; with AI on, Koji phrases over the identical diagnosis (one source of truth for "why wrong").
~ 7-skill taxonomy (identify-sides, areas-of-squares, theorem-statement, find-hypotenuse, find-a-leg, right-triangle-test, coordinate-distance) tagged on problem steps	 Shipped	The unit that spacing, retrieval, and mastery all key off; small and bounded, matching the one-chapter course.

Decision	Status	Why
Whole-number generated problems: Infinite Practice authors only Pythagorean-triple geometry, so hypotenuses and answers come out as whole numbers (never an irrational that displays as an ugly decimal like 22.2), and "count the squares" grids stay small and countable	✓ Shipped	A <i>pedagogical-fitness</i> constraint that now lives in the prompt, not the firewall: the model is instructed to produce whole-number, triple-based problems and <i>owns</i> that it does. The earlier server/client check that deterministically rejected any non-integer answer before display was removed along with the rest of the generation round-trip (see B.7 and SPOV 2's scoped exception); only crash-prevention shape-validation and a deterministic fallback remain. The model authors and certifies; the engine no longer re-derives.
Batch-generated, cached Infinite Practice: author a batch (~5) in parallel, each call given a distinct <i>variety seed</i>, tuned to the learner's current state/mastery, cache it in Firestore (<code>users/{uid}/practiceCache</code>), serve from cache with prefetching, regenerate only when the learner's state changes (difficulty-based invalidation)	✓ Shipped	An efficiency/architecture call: batching plus a warm, prefetched cache means far fewer model calls and lower latency and cost than authoring one problem at a time, and the per-call variety seed keeps a batch from collapsing onto one triple. Cached problems are model-authored and trusted, not firewall-verified (the generation round-trip was removed; see B.7 and SPOV 2's scoped exception); the path still stays AI-off-safe, with AI off the generator degrades gracefully and the fixed lesson path carries the course (SPOV 4).

Decision	Status	Why
Model-authored Infinite Practice generation: a strong reasoning model (gpt-5.5 at xhigh reasoning) authors each complete problem and owns its correctness; all authoring rules (whole-number answers, correct math, per-kind concept variety, figure-template choice) live in the prompt, with a distinct per-call variety seed; the deterministic correctness round-trip is removed, leaving only crash-prevention shape-validation and a deterministic fallback	✓ Shipped	A deliberate, scoped reversal of SPOV 2 for the <i>generated-practice path only</i> (the firewall stays absolute for the core teaching engine): a wrong supplemental practice problem is low-stakes, and re-deriving every variant proved too rigid, forcing repetitive triples and blocking variety. Honest bound: a generated problem can now occasionally be wrong, by design, and can recur across sessions (no server gating, no cross-session memory). Full argument in B.7 .
Per-learner FSRS parameter training	✗ Skipped	Defaults already beat SM-2 for ~99.5% of users; training needs a review history we won't have on day one and adds ML infra not yet worth it. (<i>SPOV 7, stated honestly.</i>)
Interleaving as a standalone headline mode	✗ Skipped	Interleaving is real, but it rides the cumulative/mixed review sessions; a separate "interleave" surface would be décor.
AI-authored feedback / pedagogy	✗ Skipped	The diagnosis is deterministic and the model only phrases; authoring pedagogy stays out of scope (SPOV 2/4/9).
Full cross-lesson adaptive sequencing beyond mastery + SR	»» Deferred	The fixed hand-authored path still owns order; sequencing the <i>curriculum itself</i> is a later bet.
Multi-level scaling of the taxonomy / second chapter	»» Deferred	The model generalizes, but the app keeps depth at one level, depth over breadth, the project's governing rule.

B.4 Code analysis: AI-generated vs hand-written

Same posture as the engine and AI layer: **~100% AI-written code / 100% human-decided constraints**. The learning-science layer's leverage was, again, in framing the problem so the swarm couldn't ship something wrong, here that meant the *locked spec* (the four techniques, the additive/AI-off rule, the **outcome→grade mapping**, the **"mastery**

= **survives a spaced review**" redefinition, and the decision to ship FSRs **defaults**) plus a single new validation bar: **unit-test the FSRs port against the open-source reference**. The notable new artifact is that FSRs-6 TypeScript port, AI-written from the open-source spec and AI-validated against it under human-specified test criteria. As with the rest of the app, the human contribution was direction and judgment, not keystrokes.

B.5 Key learnings (the seeds of SPOVs 6–9)

- **Mastery must be durable.** The hardest pedagogy call was refusing to count a one-time pass as mastery; tying the gate to *surviving a spaced review* is what makes the signal mean something. → **SPOV 6**.
- **A modeled scheduler beats a vibe, and honest defaults beat a hand-tuned multiplier.** The value of FSRs isn't a few percent on seven skills; it's a real forgetting-curve model and a clean lapse mapping, shipped with defaults and *said out loud*. → **SPOV 7**.
- **Make reviews retrieval, not recognition.** A review that feels easy is usually the wrong one; the system has to choose the harder, generative, cumulative review for the learner. → **SPOV 8**.
- **Put the diagnosis below the AI line.** Moving the misconception classifier into the deterministic layer made the AI-off path as sharp as AI-on, and gave the model one source of truth to phrase over. → **SPOV 9**.
- **Additive-on-primitives pays a second dividend.** Because the whole layer rode `recordStep` + the engine + the content model, the learning-science features added *no* AI dependency and stayed AI-off-safe by construction, the original additive architecture earned its keep twice.
- **Porting a benchmarked open-source model beats hand-rolling intervals.** Cloning Anki / `fsrs-rs` and unit-testing the port was faster *and* more correct than inventing a schedule, and it made the honesty (defaults, not trained; directional, not measured) precise rather than hand-wavy.

B.6 Mastery, once earned, is permanent (the badge is decoupled from the scheduler)

The mastery system does two jobs that the first implementation (`applyOutcome` in `src/lib/learning/mastery.ts`) quietly fused into one. The first is a **retention mechanism**: the FSRs scheduler, whose job is to keep resurfacing a skill for as long as the learner still needs reinforcement. The second is an **achievement signal**: the visible

"Mastered" badge, a motivational marker of something the learner earned. The original code let a single lapse (a wrong or Koji-assisted answer, which the outcome map grades as FSRS "Again") do *both* jobs at once: it collapsed the skill's stability (correct) *and* demoted the skill from "Mastered" back to "Provisional" (the mistake). The app keeps the first effect and removes the second, so a skill that has reached Mastered now stays mastered permanently. The earning bar is untouched and the scheduler is untouched; only the demotion is gone. This refines the durable, gated mastery row of B.3, changing how mastery can be *lost*, not how it is *earned*, and like the rest of the layer it lives in the deterministic path, so it holds with the AI flag off.

The counter-view, stated honestly and at full strength. The purist durable-mastery position argues the opposite of what we shipped, and it is a serious case. If "mastery" is meant to track *current* retention, then it must be **continuously re-verified**, and the badge should be revoked the instant the skill lapses, because a learner who can no longer produce the skill unaided is, by definition, no longer demonstrating mastery of it. On that reading a permanent badge can **overstate the learner's present retention**, the very learning-versus-performance gap (Soderstrom & Bjork, 2015) that SPOV 6 exists to respect, and a mastery meter that only ever rises risks decaying into a participation trophy, decorative rather than diagnostic.

The spiky case for decoupling the badge from the scheduler. That argument is right that the retention *truth* must keep up with reality; it is wrong that the *badge* is where that truth should live. So the app splits the two jobs and gives each to the layer that does it best. The **earning bar stays demanding**: a skill becomes Mastered only after a first-try-correct demonstration *and* surviving at least one genuinely due spaced review (SPOV 6, SPOV 7), so mastery is still hard to earn. The **scheduler still tells the truth**: a later lapse still collapses the skill's FSRS stability, so the skill resurfaces for spaced reinforcement soon (the retention mechanism is exactly as aggressive as it was before). But the **badge is never stripped**, for two reasons. First, a single lapse is *weak evidence*: it cannot separate a momentary slip (a misread, a mistyped value, a Koji-assisted nudge on an off day) from genuine forgetting, and the scheduler already covers the only case that matters by bringing the skill back regardless. Second, demoting the learner on that weak signal is not free; it is *behaviorally costly*. By loss aversion (Kahneman & Tversky, 1979), losing an earned "Mastered" is felt far more sharply than the original gain was, so a demotion does motivational damage out of all proportion to the information it carries; and stripping a

hard-won competence marker undercuts the **competence need** that self-determination theory names as a pillar of intrinsic motivation (Ryan & Deci, 2000). The badge's whole job is to certify "you built something real," and revoking it on one slip says the reverse.

The honest bound. The decision's cost is exactly the counter-view's best point, conceded plainly: the "Mastered" badge is an *achievement* record, not a *live* retention readout, so a learner who lapsed yesterday still wears it today. We accept that gap because nothing in the system treats the badge as a retention measure: the live readout is the scheduler's per-skill D/S/R and the "Reviews due (N)" hub, which *do* react to a lapse immediately, so a lapsed-but-still-badged skill is already queued for reinforcement. The badge says "you earned this"; the scheduler says "practice it now." Keeping those two statements separate, an achievement that persists and a retention estimate that moves, is the whole point of the change.

B.7 Generated practice trusts the model: a scoped exception to the firewall

Infinite Practice generation is now **model-authored**. A strong reasoning model, **gpt-5.5** (at *xhigh* reasoning), writes the *complete* problem, its prompt, givens, answer, and figure choice, and **owns its correctness**. Every authoring constraint the deterministic layer used to enforce now lives in the **prompt**: answers must be whole numbers, the math must be right, problems must vary, and any figure must reference one of the hand-built visual templates. The deterministic *verification* firewall that once stood on this path, the step that re-derived each generated answer and rejected anything non-integer or non-recomputable, has been **removed for generation**; what remains is only **crash-prevention shape-validation** (is this object well-formed enough to render without throwing?) and a **deterministic fallback** (if the model is off, slow, or returns something malformed, the app serves a safe, hand-shaped problem instead). Variety, too, is now prompt-driven rather than verifier-driven: each of the parallel batch calls is handed a distinct "**variety seed**," and each problem *kind* carries its own **concept-variety constraints**, so a batch of "find the hypotenuse" items isn't secretly five reskins of the same 3-4-5. There is **no server gating** of what the model returns and **no cross-session memory** of what it has already produced. This is a deliberate, *scoped* reversal of **SPOV 2** for the generated-practice path **only**; for the core teaching engine, the firewall stays exactly as absolute as SPOV 2 describes.

The counter-view, stated honestly and at full strength. The firewall purist, the SPOV 2 hardliner, has the stronger-sounding case, and it deserves full strength. A Pythagorean answer is the *single most checkable thing in the entire product*: re-deriving $\sqrt{a^2+b^2}$ and confirming it is an integer costs microseconds and essentially cannot fail. If

correctness is that cheap to verify, *why would you ever not?* Removing the check buys nothing on the cases it would have caught and re-exposes the learner to exactly the failure mode this BrainLift was built to eliminate, a confidently-wrong math problem, presented by a *teaching* product, with the app's authority behind it. And the asymmetry SPOV 2 leans on does not obviously evaporate just because the surface is labeled "practice": a learner drilling a wrong problem can encode the wrong procedure, lose trust the moment they catch the error, or, worse, *not* catch it. On this reading, "trust the model because it's only practice" is precisely the rationalization the firewall exists to refuse, and a strong reasoning model is still a *probabilistic* one: a lower error rate is not a zero error rate.

The spiky case for trusting the model on the practice path. That argument is right about the core engine and wrong about its scope. The firewall's job was never "verify everything checkable for its own sake"; it was "never let the model decide what the app *teaches as true*." Generated practice sits on the far side of that line: it is *supplemental*, the course teaches completely without it (SPOV 4), and a single wrong practice item is a categorically cheaper failure than a single wrong *lesson*, the one the learner is told to trust. Against that low, bounded downside sits a real, recurring cost the deterministic verifier was actually imposing: re-deriving and integer-gating *every* generated variant turned out to be **too rigid in practice**. To survive the round-trip, the generator kept retreating to the same small stable of memorized Pythagorean triples, and the integer gate rejected most of the interesting near-misses, so "infinite" practice quietly collapsed toward a dozen reskinned 3-4-5s and 5-12-13s, the precise *opposite* of the variety that makes practice worth doing (and that SPOV 8's desirable-difficulty argument depends on). So the app makes the bet explicitly: hand authoring to a model strong enough to be trusted *when constrained by a careful prompt*, spend the recovered freedom on genuine variety (variety seeds plus per-kind concept constraints), and accept a small, named correctness risk on the one path whose stakes are low enough to carry it. The firewall is not abandoned; it is **scoped to where correctness is load-bearing**, and a calculated trust takes its place where it isn't.

The honest bound. The decision's cost is the counter-view's best point, conceded plainly: because nothing re-derives a generated answer anymore, an occasional generated practice problem **can now be wrong, by design**, and we ship that risk knowingly rather than pretend a strong model is infallible. Two further limits are equally honest. First, **variety is still bounded**, not literally infinite: each difficulty band has a fixed leg range, so the space of distinct problems a learner can meet is finite, and at the easy end it is small. Second, with **no cross-session memory and no server gating**, the same problem can recur across sessions; the generator does not remember what any given learner has already seen. We

accept all three because the blast radius is contained: this is *practice*, not *teaching*; the core engine still grades the learner's own answer deterministically; the corrective loop (SPOV 6) and Koji (when on) catch a learner who is genuinely confused; and if the model is ever unavailable, the deterministic fallback and the fixed lesson path are right there underneath (SPOV 4). The firewall still guards the place a wrong answer would be disqualifying; it simply no longer polices the place a wrong answer is merely, and recoverably, annoying.